

# NATIONAL FOOD BASED DIETARY GUIDELINES

LIBRARY  
Khan Abdul Aziz Khan  
National Institute of Health  
ISLAMABAD



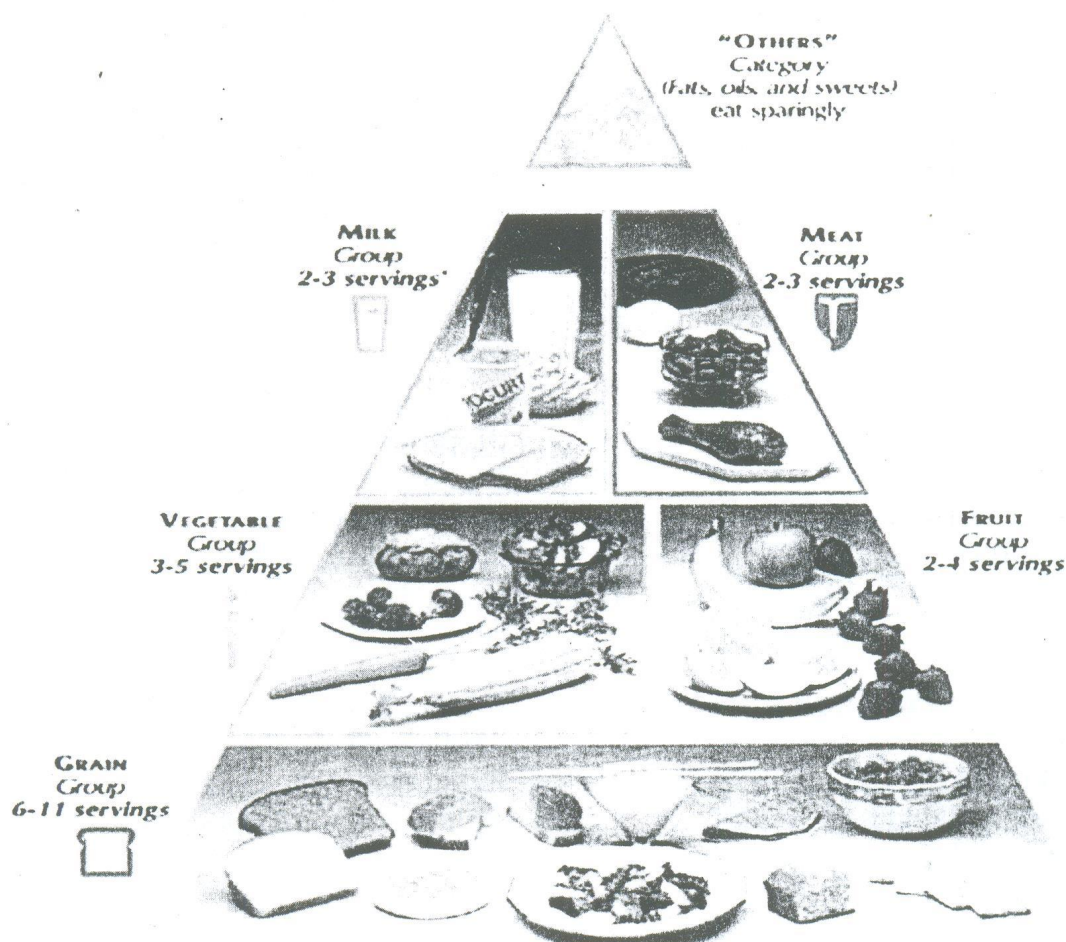
**Nutrition Division**  
National Institute of Health  
Islamabad



**Nutrition Wing**  
Ministry of Health  
Government of Pakistan  
Islamabad

LIBRARY  
KANSAS DISTRICT ALL AMERICAN  
ELEMENTARY COLLEGE of Home Economics  
KABACHI.

# Food Pyramid



Don: 375  
641.1.  
PAK.

## Prepared by

**Dr. Riffat Aysha Anis**

Principal Scientific Officer/Acting Chief  
Nutrition Division  
National Institute of Health  
Islamabad

**Prof. Dr. Parveen Liaqat**

Chairperson  
Dept. of Home & Health sciences  
Allama Iqbal Open University  
Islamabad

**Dr. Amjad Ali**

Scientific Officer  
National Institute of Health  
Islamabad

**Dr. Zahid Larik**

Deputy Director General  
Nutrition Wing  
Ministry of Health  
Government of Pakistan  
Islamabad

**Shahid Riaz**

Scientific Officer  
Nutrition Division  
National Institute of Health  
Islamabad

**Dr. Muhammad Haroon**

**Jehangir Khan**  
Deputy Director General (PHC)  
Ministry of Health  
Government of Pakistan  
Islamabad

**Dr. Farzana Chaudhary**

Chief  
DC&TMD  
National Institute of Health  
Islamabad

**Lt. Gen (R)**

**Prof. Dr. K.A. Karamat**  
Executive Director  
National Institute of Health  
Islamabad

# Table of Contents

Foreword

Executive Summary

## Part I:

|      |                                   |   |
|------|-----------------------------------|---|
| 1.   | Introduction                      | 1 |
| 2.   | Rationale                         | 2 |
| 3.   | Objectives                        | 2 |
| 4.   | Methodology                       | 2 |
| 4.1. | Pre-testing of Dietary Guidelines | 3 |
| 4.2. | Steps for Preparing Guidelines    | 4 |

## Part II:

|        |                                                   |    |
|--------|---------------------------------------------------|----|
| 5.     | Results / Outcome                                 | 4  |
| 5.1.   | Energy Requirements                               | 4  |
| 5.2.   | Food Guide Pyramid                                | 4  |
| 5.2.1. | Food Groups                                       | 5  |
| 5.3.   | Meal Patterns                                     | 5  |
| 5.4.   | Food Serving (Daily)                              | 6  |
| 5.5.   | Sample Menu (One day)                             | 7  |
| 5.6.   | Food Based dietary Guideline (Children 0-5 Years) | 8  |
| 5.6.1. | Daily Dietary Requirements                        | 8  |
| 5.6.2. | Food Serving                                      | 8  |
| 5.6.3. | Infants (0-6 Months)                              | 8  |
| 5.6.4. | Infants (6-12 Months)                             | 9  |
| 5.6.5. | Children (1-5 Years)                              | 10 |

## Part III:

|      |                      |    |
|------|----------------------|----|
| 6.   | Public Messages      | 11 |
| 6.1. | Adults               | 11 |
| 6.2. | Children under Five  | 11 |
| 7.   | Nutritional Labeling | 12 |
| 8.   | Conclusion           | 12 |
| 9.   | Bibliography         | 13 |

## Part IV:

|     |           |    |
|-----|-----------|----|
| 10. | Annexures | 14 |
|-----|-----------|----|

## Foreword

Food Based Dietary Guidelines are required to provide dietary information for the policymakers, nutrition educators and health care providers in view of the other factors that influence the health status and are more so of greater importance in the developing countries. The major emphasis is to educate the consumers in promotion of health and well being and reducing the risk of chronic non-communicable diseases.

Recognizing the importance of good nutrition and exercise on the health and therefore the need for the National Food Based Dietary Guidelines (NFBDGs) this responsibility was taken over by the Nutrition Division, National Institute of Health, Islamabad, in view of monitoring and surveillance of the nutrition situation in the country. This document is a practical action / strategy towards the reduction of malnutrition among the general population.

The drafting of these guidelines required inputs from stakeholders which was obtained by arranging workshops / meeting. The experts from the relevant disciplines contributed valuably in developing the NFBDG's and their support is highly admirable.

Development of the NFBDG's have been possible through the financial support from the Research Component under the project "Improvement of Nutrition through PHC, Nutrition Education and Awareness Program" provided by the Ministry of Health, and is greatly acknowledged.

Hopefully this first effort will serve as a stepping-stone for changing the dietary habits for the betterment of the people's health. To assist us in improving these guidelines any ideas and suggestions forthcoming from any quarter will be welcomed.

**Maj. Gen (R)**  
**M.I. Burney**  
Scientist Emeritus  
HI (M) SI (M)  
National Institute of Health  
Islamabad

## **Executive Summary**

Food-based dietary guidelines (FBDGs) in this report are the expression of the principles of nutrition education with special emphasis to foods. The general public intends them for use, as they provide science-based evidence to promote health and reduce the risk of chronic diseases like hypertension, cardio vascular diseases, diabetes etc through diet and physical activity. By following the dietary guidelines one can enjoy better health and minimize the chance of getting non-communicable diseases.

In Pakistan the major causes of morbidity and mortality are related to poor diet and sedentary life style. Furthermore, poor diet and physical activity results in energy imbalance (more calories consumed than exhausted), and contributes to increase in overweight and obesity.

To enhance the health of the population there is a need for developing food- based strategies that are locally appropriate, as FBDGs will vary among population groups in the provinces of the country. Considering diet and health relationships in a culturally sensitive way and providing consumers with easier ways to make healthy food choices is the foundation of the food based dietary guidelines approach.

A balanced diet is not the only component of a healthy lifestyle. Although this document has laid major focus on diet, but it is imperative to integrate this with other messages related to health promotion along with the dietary guidelines. Through these FBDGs the main aim is to disseminate information to a consumer in familiar food related terms of food rather than of nutrients.

# National Food Based Dietary Guidelines

## 1. Introduction

Worldwide malnutrition is a major public health problem of the developing countries. In view of this situation, which is a matter of concern for the international health organizations, an International Conference on Nutrition was held in Rome in 1992. One recommendation emanating from the deliberations calls attention of the member countries to encourage the use of strategies and actions for improved nutritional well-being and food consumption throughout the world. The purpose of this consultation was therefore, to establish the scientific basis for developing and using food-based dietary guidelines for the public, taking into account the realities and customs of each country.

Food Based Dietary Guidelines (NFBBDGs) are an instrument and an expression of food and nutrition policy and should be based directly on diet and disease relationships with particular relevance to the population of the country and/or group (s), such as children, pregnant and lactating women, and the elderly. Their primary purpose is to educate health-care professionals and consumers about health promotion and prevention of disease.

Proper nutrition from conception through all stages of the life cycle is essential for leading a healthy, productive, and high-quality life. Regular physical activity and proper diet, which are components of a healthy life-style, prevent many diseases; promote good growth and high functional capacity thereby contributing to full physical and intellectual development.

Many factors, which include knowledge, customs, culture, the availability of the food, and economic access to eatables, influence a family's food selection and consumption. These factors, in turn, can be conceptually influenced through advertising, by changing the trends or fads, social and demographic patterns that together determine the eating habits of the individuals.

Essentially the guidelines are directed towards the general population or healthy segments to promote health and reduce the risk of nutritional disorders. The guidelines are based on the nutritional and energy requirements and dietary recommendations, however, for their preparation, it is important to recognize the anthropological, cultural, educational, social and economic factors that are closely linked to the eating habits and life style of the individuals. Dietary guidelines for persons suffering from disease(s) require formulation of specific disease related manual.

Recognizing the need for preparing national food based dietary guidelines a two days workshop was held on 15-16 May 2003 in the NIH, Islamabad. Expert Nutritionists, Food Scientists, Food Technologists, Doctors and Public Health experts attended the workshop (Annex-I&II). Following the recommendations of this workshop a Task Force was constituted and a second workshop was conducted on March 28-29, 2005. The recommendation of the second workshop resulted into drafting of food based dietary guidelines (Annex - III&IV) for normal healthy adults and children aged 0-5 years to be followed by those for other vulnerable groups of population.

## **2. Rationale**

There was good scientific evidence that a dietary pattern that is the daily combination of foods and beverages have specific results in terms of health and disease. For example, a diet may be apparently adequate in all other ways but still be deficient in vitamin A or iron, thereby leading to xerophthalmia or anemia. Conversely, a diet high in saturated fats and energy is known, on a population basis, to lead to an increased likelihood of coronary heart disease.

The reasons for developing and using FBDGs are numerous and apparent:

1. Foods constitute diets, and are more than just a collection of nutrients.
2. Nutrients interact differently when presented as foods.
3. Methods of food processing, preparation and cooking influence the nutritional value of foods.
4. Evidence that specific dietary patterns are associated with a reduced risk of specific diseases. Fruits and vegetables have positive impact on cancer.
5. Food and diets have cultural, ethnic, social and family aspects.

## **3. Objectives**

The following criteria has been set for preparing the FBDGs: -

- To establish the scientific basis for developing and using FBDGs
- To improve the food consumption patterns
- To improve the health and nutritional well being of individuals and the population at large.

## **4. Methodology**

Extensive literature and dietary guidelines of different countries were reviewed and WHO/FAO dietary recommendations for a healthy lifestyle were used as the guiding principles for developing the NFBGDs:

- Eat less sugar
- Eat less fat and deep fried foods
- Eat more fiber and starchy foods
- Eat less salty food
- Eat a variety of foods
- Enjoy your food
- Eat more seasonal fruit and vegetables
- Eat the right amount to maintain a healthy weight

While developing the dietary guidelines, emphasis was laid on the following issues:

- Would a single set of guidelines suffice or was a multiple set required, e.g. for urban and rural populations?
- Should the guidelines include children, infants and other vulnerable groups?
- Should branded, packaged foods be included in the food groups?
- Should the guidelines address issues of food and water safety and sanitation, including the safety of complementary foods for young children?
- Should water and other drinks be included in the guidelines?
- Should the guidelines address weight, exercise and other lifestyle factors?
- Should the guidelines be compatible with the potential food supply?
- Are the guidelines environmentally sustainable?
- Should the guidelines be made available in the different languages used in the country?
- How many food groups are needed, and how will they be named?
- Accommodation of the wide range of nutrients between individual foods in food groups
- How can allowance be made for the wide range of energy requirements between different subgroups and individuals?
- How can recommendations be expressed on specific food components in the food groups such as protein, fats, fiber, salt and sugar?
- Meal patterns

Other basic questions that were considered important were also addressed and are placed at Annex-A.

#### **4.1 Pre-testing of National Food Based Dietary Guidelines**

The dietary guidelines developed were subjected to pre-testing prior to finalization, in all the provinces of Pakistan including AJK. A questionnaire was developed, which was used as an instrument for field-testing of the dietary guidelines and public messages. The questionnaire along with the results of pre-testing is attached (Annex - C and D).

## 4.2 Steps for Preparing Guidelines

The various steps involved in the preparation of food based dietary guidelines are illustrated in the flow diagram (Annex-B).

## 5. Results / Outcome

National Food Based Dietary Guidelines have been developed for the following age groups:

- **Normal Healthy Pakistani Adult**  
(Age 15-45 Years)
- **Normal Healthy Pakistani Child**  
(Age 0-5 Years)

### 5.1 Energy and Protein Requirements of Normal Healthy Adult:

**Male:** 2550 Kcal - Protein 95 g (15% of total Kcal)  
**Female:** 2160 Kcal - Protein 81 g (15% of total Kcal)

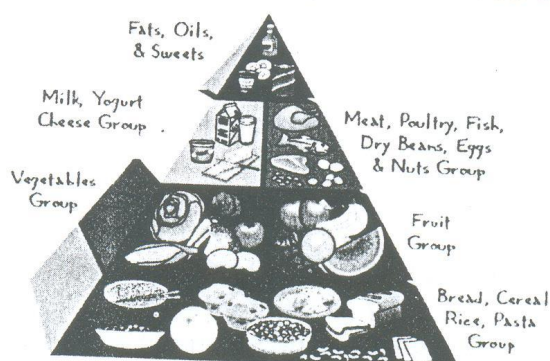
#### Caloric Value of Foods

- Carbohydrate 4 kcal per g.
- Protein 4 kcal per g.
- Fat 9 kcal per g.

**Reference:** (Food Composition Table for Pakistan 2001)

### 5.2 Food Guide Pyramid

The USA food pyramid was taken as a guideline to develop the national food pyramid. The modifications done were in line with the food items consumed by the Pakistani population based on the following food groups:



### **5.2.1 Food Groups**

- Cereal and cereal products and starchy foods (Potatoes, sweet potatoes, pasta etc.)
- All vegetables and fresh legumes
- Dry and fresh fruits
- Milk and milk products
- Meat, egg, beans and lentils
- Edible oils and fats, sugar, salt

### **5.3 Meal Pattern**

Meal Patterns did not vary between the four provinces of the country, as roti / chappati, paratha and rice are generally consumed with a variety of dishes known as curry, prepared with pulses, vegetable cooked with or without meat.

Traditional curry contains large amounts of oil (animal or vegetable origin); this trend is gradually declining due to awareness. However, the use of edible oil is being preferred over animal fat.

Meal consumption patterns are affected to some extent by seasonal changes. In summer more milk/ milk products are consumed, whereas in winter people eat relatively more meat, fish, egg and milk based sweet dishes. There are major seasonal variations in vegetable and fruit consumption.

## 5.4 Daily Food Servings

Culturally food servings in Pakistan are large, medium and small depending on the socio-economic norms / status. The daily food servings for Pakistan population were tailored keeping in view the International Standard of Serving for normal healthy Pakistani adult. These recommendations were based on the provincial acceptability and considering the Protein Energy Malnutrition (PEM) and micronutrient (iron, iodine and vitamin A) deficiency diseases. The suggested Pakistani daily food servings are tabulated as under:

| Food Group                                                                        | Daily Servings | 1 Serving Size                                                                                                                                           |
|-----------------------------------------------------------------------------------|----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|
| Milk and Milk Products                                                            | 2-3            | 1 Cup* Milk or Yogurt<br>1/2 Cup Indigenous (Cottage) Cheese<br>1/4 Cup Plain Ice cream                                                                  |
| Meat, Egg, Beans, Pulses and Nuts                                                 | 2-3            | 50 g cooked Lean Meat, Poultry or Fish<br>1 Egg / 2 Egg Whites<br>1/2 Cup cooked Legumes (Dried Beans or Pulses/Dal)<br>30 g Nuts                        |
| All Vegetables and Fresh Legumes Group                                            | 3-5            | 1/2 Cup cooked Vegetables.<br>1/2 Cup Salad<br>1 Cup Raw Leafy Vegetables                                                                                |
| Fruit Group (All fresh and dried)                                                 | 2-4            | 1 Medium Size Fruit (about 100 g)<br>1/4 Cup Dried Fruit<br>1/2 - 3/4 Cup Fruit Juice                                                                    |
| Cereal and Cereal Products and Starchy Foods (potatoes, sweet potato, pasta etc.) | 5-9            | 2 Slice Bread<br>1 Small Chapatti<br>4 Small Rusk<br>1 Cup Cooked Cereal<br>1 Cup Cooked Rice<br>1 Cup Noodles                                           |
| Edible Fats and Oils Sugars and Salt                                              | 2-4            | 1 Tablespoon Oil (10 g)<br>1 Tablespoon Butter/Margarine<br>1 Tablespoon Mayonnaise<br>1 Tablespoon Salad Dressing<br>Use Sugar sparingly<br>Honey 10 g. |

\*1 Cup = 250 ml

## 5.5 Sample Menu (One-Day)

One-day sample menu has been prepared, that is based on the Pakistani food pyramid and daily food servings with the objective to provide a guideline for people to plan diet according to their daily requirements and within their resources.

### Servings of Various Food Groups Contributing to Sample Menu

|                  |                |
|------------------|----------------|
| Milk Group:      | 2 Servings     |
| Meat Group:      | 1-1/2 Servings |
| Vegetable Group: | 5 Servings     |
| Fruit Group:     | 1 Servings     |
| Cereal Group:    | 6-7 Servings   |

**Breakfast:** 1 Chapatti / Paratha / Bread / Bun  
1/2 Cup Vegetable Bhujia / Acchar / Egg / Curd  
1 Cup Beverages (Tea, Khawa, Lassi)

**Midmorning:** 1 Rusk / Biscuit  
1 Cup Tea / Lassi

**Lunch:** 3/4 Cup Dal  
1-Cup Vegetable Bhujia  
1-Cup Pakistani Salad  
2 Chapattis / Rice  
2-3 Tablespoons Chutney/ Raita  
1 Fruit (seasonal)

**Evening Tea:** 1 Cup Tea / Lassi/ Lemonade/ Sattu

**Dinner:** 3/4 Cup Dal / Beans  
1-Cup Vegetable Bhujia  
1/2 Cup Salad  
2-3 Chapattis / Rice  
2-3 Tablespoons Chutney/ Raita

### Caloric Calculation of Sample Menu

- Calories: 2300 (Approx. for both man / woman weighing 55 kg)
- Protein: 80 g (Approx. for both man and woman)
- Cost: Rupees 50-60/- Approx.

## 5.6 Food Based Dietary Guidelines for Children (0-5 years)

### 5.6.1 Energy and Protein Requirements (Daily)

| Age Group    | Body Weight (Kg)         | Caloric Requirements (Kcal) | Proteins Requirements (g) |
|--------------|--------------------------|-----------------------------|---------------------------|
| 0-6 months   | Exclusive Breast Feeding |                             |                           |
| 6 –12 months | 8.5                      | 950                         | 14                        |
| 1-5 years    | 11.5 - 19                | 1350 - 1600                 | 22 - 26                   |

(FAO – 1997)

### 5.6.2 Food Servings for Children

| Food Groups                                                                           | Daily Serving | 1 Serving Size                                                                                                        |
|---------------------------------------------------------------------------------------|---------------|-----------------------------------------------------------------------------------------------------------------------|
| Cereals and Cereal Products and Starchy Foods (Potatoes, Sweet Potatoes, Pastry etc). | 5 – 6         | ½ Chappati / Roti / Naan<br>1 Slice of Bread<br>½ Cup Cooked Rice<br>½ Cup Cooked Cereal<br>½ Cup Ready to Eat Cereal |
| Meat, Egg, Beans                                                                      | 2 – 3         | 100 g Cooked Lean Meat<br>100 g of Poultry / Fish<br>1 Egg<br>½ Cup Cooked Beans                                      |
| All Vegetables and Fruit Legumes Group                                                | 2 – 3         | ½ Cup Cooked Vegetables<br>½ Cup Raw Vegetables<br>1 Cup Leafy Raw Vegetables                                         |
| Fruit Group                                                                           | 2 – 3         | 1 Apple / Banana / Orange<br>½ Cup Seasonal Fruit<br>½ Cup Fruit Juice                                                |
| Milk and Milk Products                                                                | 2 – 4         | 1 Cup Milk<br>1 Cup Yogurt<br>50 g Cheese                                                                             |

### 5.6.3 Infants (0-6 months)

Human milk provides the infants with most of the nutrients needed for optimal growth and development. It is recognized as an ideal food for infants / children, except in few cases where specific contraindications exist. **Exclusive breast-feeding** is recommended from birth till 6 months of age.

#### 5.6.4 Infants (6-12 months)

By the age of 6-12 months, ability to swallow non-liquid foods has been established. Introduce semi-solid foods at the age of 6 months along with the breast milk.

At the age of six months start with one semi solid food in small quantity (1-2 spoons) and introduce other type of foods gradually and regularly twice a day. Continue breast-feeding and increase the frequency and quantity of complementary foods with age. One-day sample menu for 6-12 months children is:

##### **Sample Menu for 6-12 Months**

###### **Breakfast**

Semi Liquid Porridge/ Suji / Custard /Kheer ¼ Cup garnished with Desi ghee/ Butter / Margarine / Oil  
Milk (½ Cup)

###### **Mid-Morning Snack**

One Biscuit / Rusk  
Milk (½ Cup)

###### **Lunch**

Rice with Yogurt / Lentil / Cooked in Meat Broth (Yakhni) / Mince Meat ¼th Cup  
Add Mashed Green Leafy Vegetables (¼ Cup)  
Mashed Banana / Mango / Peach or any other Seasonal Fruit / (½ Cup)

###### **Mid - Afternoon Snack:**

Apple / Seasonal fruit juice (¼ Cup)  
One Biscuit / Rusk

###### **Dinner**

Mashed Chicken / Fish/ Minced Meat ((¼ Cup)  
Mashed Potato (¼ Cup)  
Green Leafy Vegetables / Peas / Beans / Lentil.  
Mashed Pear / Apple / any Seasonal Fruit (3 Table spoons)  
Milk (½ Cup)

###### **Bed Time Snack**

Biscuit / Rusk (1)

Milk (1 Cup)

#### **5.6.5 Children (1-5 years)**

The pre-school diet is designed to promote growth and development; diet should be adapted to meet preferences and needs of the children. When a variety of foods are consumed, the diet meets the Recommended Dietary Allowances (RDA) for that age group.

Serve three meals and 2-3 nutritious snacks daily. Between-meals snacks should be planned so that they do not interfere with appetite at mealtimes, but snacks are part of the total food intake.

Children usually prefer mild flavors and simply prepared foods that are soft in texture and easy to chew. Spicy and fried foods may not be tolerated. Children should be encouraged but never forced to eat. Food should never be used as a discipline or reward. Sample menu for children 1-5 years is given below:

#### **Sample Menu (1-5 years)**

##### **Break Fast**

Milk (1 Cup)

Cooked Cereal / ready to eat Cereal (½ Cup)  
or Chapatti /Nan/ Roti (½)

Egg (1)

##### **Mid – Morning Snack**

Banana / Orange / Apple (1)

Biscuit (1)

##### **Lunch**

Cooked Rice / Cooked Pasta (½ Cup)

Chapatti / Roti / Nan ½

Cooked Lean Meat /Poultry / Fish or ½ cup of Cooked Beans

Cooked / Raw Vegetable (½ Cup)/ Leafy Green Vegetable (1 Cup)

##### **Mid – Afternoon Snack**

Fried / Boiled Potato / Vegetable garnished with Margarine / Oil /  
Butter (½ Cup)

Biscuit / Rusk (1)

### **Dinner**

Cooked Vegetable ½ Cup or Leafy Green Vegetable (1 Cup)  
Chappati / Roti /Naan ½ or Slice of Bread (1).  
Banana / Orange / Apple (1) or Fruit juice or any Seasonal Fruit (½ Cup)

### **Bed Time Snack**

Milk / or Yogurt (1 Cup)  
Biscuit / Rusk (1)

LIBRARY  
2012  
2012  
2012

## **6. Public Messages**

Draft messages prepared in advance for Adults and Children under five were deliberated upon and discussed by the participants and approved as:

### **6.1 Messages for Adults:**

|                                                                                                                     |
|---------------------------------------------------------------------------------------------------------------------|
| 1. Eat a variety of foods at each meal.                                                                             |
| 2. Drink eight or more glasses of water daily, to avoid dehydration especially in warm climate.                     |
| 3. Cut down fried foods and sweet meats.                                                                            |
| 4. Eat vegetables (Dark Green Leafy Vegetables) and legume-based dishes. Add seasonal fruits frequently.            |
| 5. Select foods low in fat and sugar.                                                                               |
| 6. Prefer transparent vegetable oil for cooking.                                                                    |
| 7. Include a cup of milk or milk product in your daily diet.                                                        |
| 8. Balance your daily diet with your daily physical activity.                                                       |
| 9. Use less salt in the daily diet and limit traditional dishes /foods that are heavily preserved /pickled in salt. |
| 10. 10.Prefer fresh and homemade foods.                                                                             |
| 11. Physical activity on regular basis.                                                                             |

### **6.2 Messages for Children under Five:**

|                                                                                     |
|-------------------------------------------------------------------------------------|
| 1. Initiate breast-feeding soon after the childbirth within 30 minutes to one hour. |
| 2. Stick to exclusive breast-feeding up to six months of age.                       |
| 3. Introduce semi solid foods at the age of six months along with breast feeding    |
| 4. Start feeding semi solid foods twice daily at the age of six months              |
| 5. Continue breastfeeding and increase the frequency of complementary               |

|                                                                     |
|---------------------------------------------------------------------|
| foods gradually.                                                    |
| 6. Introduce new foods gradually and regularly                      |
| 7. Use clean utensils and wash hands while preparing the safe food. |
| 8. Avoid giving all types of fizzy drinks.                          |
| 9. Introduce family foods at the age of one.                        |
| 10. Switch the child to normal family meal at the age of two.       |

## 7. Nutritional Labeling

Nutritional labeling is essential to help people follow the dietary goals. Ministry of Agriculture, Fisheries and Food have produced guidelines for the nutritional labeling. According to the guidelines the food manufactures should provide complete information of the specific food to the consumer.

## 8. Conclusion

Food based dietary guidelines create awareness and encourage the healthy lifestyle within the traditional norms and cooking practices. These are expected to provide the recommended energy and protein intake to the healthy adults and children less than five years of age. Following the dietary guidelines one can enjoy better health and reduce the chance of getting non-communicable diseases. Before these dietary guidelines are disseminated to provide advice to the public they should be pre-tested on scientific basis and then disseminated through the use of mass media and other appropriate measures-qualitative and/or quantitative.

## 9. Bibliography

1. Dietary Guidelines for Americans. US Department of Health and Human Services, US Department of Agriculture, 2005.
2. FAO / WHO. International Conference on Nutrition, Major Issues for Nutrition Strategies. ICN/92/INF/7, 1992.
3. FAO/WHO Joint Consultation Report. Preparation and use of Food-Based Dietary Guidelines, WHO Technical Report Series, 880, WHO Geneva, 1998.
4. FAO 1997 Human Nutrition in Developing World. FAO Food and Nutrition Series, No 29. 4. Health for All by the Year 2000. The Finish National Strategy, Ministry of Social Affairs and Health, 1987.
5. Nancy J. Peckenpaugh and Charlotte M. Poleman Nutrition Essentials and Diet Therapy. 7<sup>th</sup> Ed. W. B. Sunder Co USA, 1995.
6. National Academy of Sciences: Recommended Dietary Allowance, Ed 10, DC, National Academy Press, 1989.
7. Peggy S. Stanfield and Y. H. Hui. Nutrition and Diet Therapy. Self-Instructional. American Dietetics Association. Modules 4<sup>th</sup> Ed. Jones and Bartlett Publishers Inc, Sudbury, Massachusetts, 2003.
8. Paul Insel, R. Elaine Turner and Don S. Rose. Discovering Nutrition. American Dietetics Association, Jones and Bartlett Publishers Inc, Sudbury, Massachusetts, 2000.
9. Sue. Rodwell William. Basic Nutrition and Diet Therapy, 10<sup>th</sup> Ed. Von Hoffman Press, Mosby- Year book, Inc, 1995.
10. WHO Dept of Nutrition for Health and Development. Complementary Feeding, Family Foods for Breastfed Children, WHO.NHD/00.1, WHO/FCH/CAH/00.6, WHO, 2000.
11. WHO/Tufts University School of Nutrition Science and Policy. Keep Fit for Life. Meeting the Nutritional Needs of Older Persons. WHO Geneva, 2002.
12. WHO/FAO Vitamin and Mineral Requirements in Human Nutrition 2<sup>nd</sup> Ed. 2004.

**Recommendations of Workshop to Develop  
National Food Based Dietary Guidelines  
15-16 May 2003**

**Mission Statement**

- “To provide comprehensive scientific basis related to food consumption, enabling the individuals to select the type, variety and quality of food for their daily requirements”. The dietary guidelines would be directed for the healthy population.

**Recommendations**

- Food group requirements for maintaining health do not essentially vary for urban and rural dwellings; therefore, a single set of dietary guidelines will suffice at the national level.
- Since ample food variety is available, the dietary guidelines and food groups would not be incompatible with the availability
- The diagrammatic presentation of the dietary guidelines would be in the shape of a pyramid.
- The dietary guidelines will be framed on broader basis to cope with the requirement of all age groups i.e. children, adults and elderly people.
- Dietary guidelines will consist of seven food groups and the task force will endorse the appropriate nomenclature.
- It will be ensured that dietary guidelines will be sustainable *viz a viz* environmental factors and availability.
- Dietary guidelines will be published bilingually in Urdu and English; however, it could be adapted to any colloquial version as needed.
- Dietary guidelines will not reflect the choices and promotion of any particular brand labeling, style and version of food item.
- National Institute of Health may constitute a task force comprising of experts from allied disciplines
- The task force will prepare micro-details of dietary guidelines and publish in a booklet / poster form after getting approval from the federal government.

**List of Participants to Develop  
National Food Based Dietary Guidelines  
National Institute of Health, Islamabad  
15-16 May 2003**

- 1. Dr. Athar Saeed Dil                      Ex-Executive Director  
National Institute of Health  
Islamabad.
- Dr. Mushtaq A.Khan                      Director Centre for Research on  
Poverty Reduction P-Block, 1st Floor  
Room, Islamabad
- Maj. Gen. (R)                      Scientist Emeritus  
M.I. Burney H.I (M) S.I.                      National Institute of Health  
Islamabad
- Mr. Muhammad Saleem                      Ex-Chief  
(Late)                      Nutrition Division  
National Institute of Health  
Islamabad
- Dr. Riffat Aysha Anis                      Senior Scientific Officer  
Nutrition Division  
NIH, Islamabad.
- Dr. Muhammad Haroon J. Khan                      Deputy Director General  
PHC Program, Ministry of Health  
Feroze Center, Blue Area,  
Islamabad.
- Dr. Farah Sabih                      Assistant Director General  
Nutrition Wing  
Ministry of Health  
Blue Area,  
Islamabad
- Muhammad Yaqoob Qureshi                      Media Advisor  
Nutrition Wing  
Ministry of Health

- Dr. Zulfiqar Khan      Technical Officer (JPRM)  
I.H.Section, Ministry of health  
Room No.133, Block-C Pak  
Secretariat,  
Islamabad.
  
- Dr. Ashfaq Ahmed      D.DG (Internal Health)  
Ministry of Health  
Islamabad.
  
- Dr. Asifa Murtaza      Head Dept. of Pediatrics, Social &  
Preventive Pediatrics & Nutrition,  
FGSH, Islamabad
  
- Lt. Col. Habib-ur-Rehman      Commandant  
Armed Forces Institute of Nutrition  
(AFIN) R.A. Bazar, Badian Road  
Lahore Cantt.
  
- Dr.Rizan Khan      Consultant Dietitian  
Shifa International Hospital  
H-8, Islamabad
  
- Dr. Pervaiz Paracha      Chairman  
Nutrition Dept  
NWFP University of  
Agriculture Peshawar
  
- Dr. Faqir Muhammad Anjum      Prof./Chairman, Food Technology  
Department, University of Agriculture,  
Faisalabad.
  
- Dr.Abid Hussain      Associate Professor/Chairman  
Dept .of Food Science & Technology  
University of Karachi  
Karachi.
  
- Dr. M. Javaid Shah Aftab      Additional Director Nutrition  
Office of the Director General  
Health, 24 Cooper Road, Lahore.

- |                         |                                                                                                    |
|-------------------------|----------------------------------------------------------------------------------------------------|
| ➤ Mrs.Saeeda Raza       | SSO Food Division<br>NARC, Chak Shahzad,<br>Islamabad.                                             |
| ➤ Mrs.Ambreen Zaka      | Dietitian<br>PNS Hafeez Hospital E-8<br>Islamabad.                                                 |
| ➤ Mrs.Bedar Bakht       | Senior Scientific Officer<br>Nutrition Division, NIH,<br>Islamabad.                                |
| ➤ Dr. Amjad Ali         | Scientific Officer<br>Nutrition Division, NIH,<br>Islamabad.                                       |
| ➤ Mr.Shahid Riaz        | Scientific Officer<br>Nutrition Division, NIH,<br>Islamabad.                                       |
| ➤ Dr.Hassan Orooj       | Health Officer<br>Directorate of Health Services,<br>CDA, F-7 Markaz<br>Islamabad.                 |
| ➤ Dr.Noor Ahmad Khan    | CIDA, Program Support Unit<br>18 Bazar Road, G-6/4,<br>P.O Box No.234,<br>Islamabad                |
| ➤ Dr. Afshan Shahid     | Lecturer Dept of Community<br>Medicine,<br>Foundation University,<br>Medical College<br>Islamabad. |
| ➤ Sophie Simon          | ACF Consultant. Main Road, G-8/3,<br>Islamabad                                                     |
| ➤ Dr. S. Moazam Hussain | Program Officer Nutrition<br>UNICEF Office<br>Islamabad                                            |

**Participants of Two Days Meeting to Develop  
National Food Based Dietary Guidelines (NFB DGs)  
28-29 March 2005**

1. Lt. Gen. (R) Prof. Dr. K.A. Karamat (Patron-in-Chief)  
Executive Director  
NIH, Islamabad  
Ph: 051-9255117  
Email: edoffice@apollo.net.pk
- Major General (Rtd) M.I. Burney  
Scientist Emeritus  
Former Executive Director, NIH, Islamabad.  
Ph: 051-5583405
- Dr. Farzana Chowdhary  
Chief, Nutrition Division, NIH, Islamabad  
Ph: 051-9255079  
Email: Shahidquetta@yahoo.com
- Prof. Dr. Perveen Liaquat (Chairperson)  
Chairperson, Department of Home and Health Sciences, AIOU, Islamabad  
Ph: 051-9250063  
Email: perveenliaquat@yahoo.com
- Dr. Riffat Aysha Anis (Secretary)  
Principal Scientific Officer, NIH, Islamabad  
Ph: 051-9255080  
Email: riffatanis@yahoo.com
- Dr. Muhammad Haroon Jahangir Khan  
DDG, Nutrition wing, Ministry of Health,  
Taimur Plaza, Blue Area, Islamabad  
Ph: 051-9202445
- Mr. Mohammad Ayub  
Acting Chief, Planning and Development Division,  
Taimur Plaza, Blue Area, Islamabad  
Ph: 051-9211348

- Prof. Dr. Rubina Hakeem  
Principal, R.L.A.K Govt. College of Home Economics, Karachi  
Ph: 0300-2391851  
Email: rubinahakeem@hotmail.com
- Dr. Parvaiz Paracha  
Dean, Human Nutrition, University of Peshawar, Peshawar  
Email: parvez\_paracha@yahoo.com
- Dr. Farah Sabih  
ADG, Nutrition wing, Ministry of Health, Taimur Chambers Islamabad  
Ph: 051-9214976  
Email: farah\_sabih@hotmail.com
- Mr. Muhammad Yaqoob Qureshi  
Media Advisor, Nutrition wing, Ministry of Health,  
Taimur Chambers, Blue Area, Islamabad  
Ph: 051-9202426
- Dr. Muhammad Zaheen,  
Deputy Director Public Health, D.G. Health Office, Khyber Road, Peshawar  
Ph: 091-9210187, 091-9210196, 091-9210231, 0333-9106341  
Email: mzaheen@brain.net.pk
- Dr. Mrs. Nusrat Ejaz  
Principal Scientific Officer, PCSIR Labs Complex, Lahore  
Ph: 042-9230688 Ext: 287  
Email: pcsir@brain.net.pk
- Dr. Javaid Aziz Awan  
Professor (Rtd), University of Agriculture, Faisalabad  
Ph: 041-8542156  
Email: awanjavaid@hotmail.com
- Dr. Mahmood Ahmad  
Ad. Director Health Services, Directorate General Health Services,  
24-Copper Road, Lahore  
Ph: 042-9203749, 0300-4314895
- Mrs. Bedar Bakht  
Microbiologist, Nutrition Division, NIH, Islamabad  
Ph: 051-9255110-14

- Mr. Shahid Riaz  
Senior Officer, Nutrition Division, NIH, Islamabad  
Ph: 0333-5162102  
Email: shahidquetta@yahoo.com
- Dr. Shimail Daud  
National Programme Officer, Vitamin A Supplementation  
The Micronutrient Initiative  
Email: sdaud@micronutrient.org
- Ms. Nadia Ishtiaq  
Lecturer (Nutrition) Department of Home and Health Sciences, AIOU, Islamabad.  
Ph: 051-9250111 Ext: 4741  
Email: nishtiaq@yahoo.com
- Mrs Noor-us –Saba  
Scientific Officer  
Nutrition Division, NIH, Islamabad  
Ph: 051-9255110-14

Following participants could not attend the meeting:

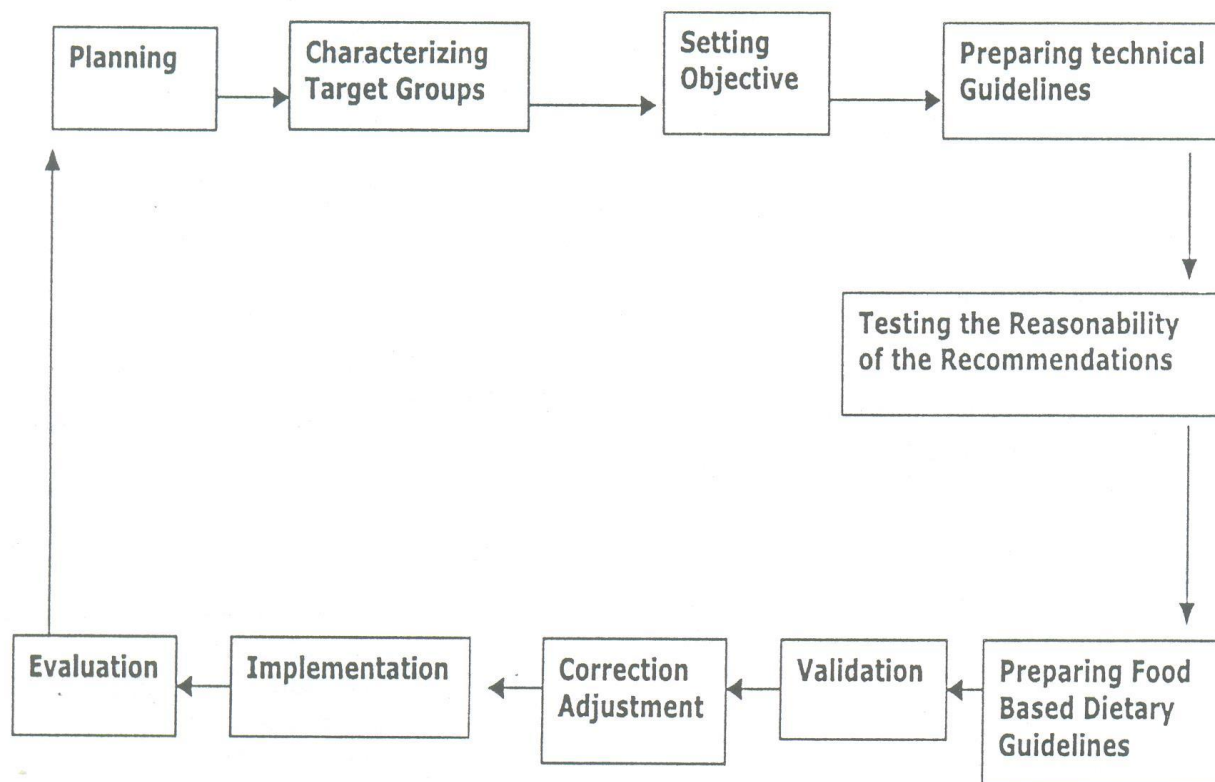
- Dr. Shabina Raza  
Deputy Director Nutrition, D.G Office, Ministry of Health  
Khyber Road  
Peshawar.
- Dr. M. Faqir Anjum  
Director Institute of Food Science and Technology  
University of Agriculture  
Faisalabad
- Prof. Dr. Mohammad Akmal Khan  
Director Monitoring and Evaluation, Tawana Pakistan  
Islamabad.
- Dr. Ali Nasar Bughti  
Deputy Director Nutrition  
Director Health services Office  
Govt. of Balochistan, Saryab Road  
Quetta

## Annex-A

The questions and the answers addressed by the participants are as:

| Questions                                                                                                                                       | Answers                                                                                                                                                                                                |
|-------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ➤ Is a single set of guidelines needed or multiple sets required, e.g. for urban and rural population?                                          | ➤ The house unanimously agreed on establishing a single set of guidelines                                                                                                                              |
| ➤ Are the guidelines to include children, infants and other vulnerable groups?                                                                  | ➤ Guidelines needs to be drawn for all the groups in a phased manner i.e. infants, preschool, school going, adolescents, adults and elderly. Further subgroups will be lactating and pregnant mothers. |
| ➤ Are the guidelines environmentally sustainable?                                                                                               | ➤ Yes, the guidelines ought to be environmentally sustainable but revision needs to be made every five years.                                                                                          |
| ➤ Should the guidelines be made available in different languages used in the country?                                                           | ➤ Yes, once tested and approved by all the authorities should be transferred in regional languages and disseminated among the target groups.                                                           |
| ➤ How many food groups are needed and how they will be named?                                                                                   | ➤ Total of six groups are needed as reflected in spectrum of magic diet prism. Standard or modified food pyramid may be used as the basis of the Pakistani diet.                                       |
| ➤ Should branded, packaged foods be included in the food groups?                                                                                | ➤ No branded, processed foods be recommended for use, for maintaining the health of a normal adult.                                                                                                    |
| ➤ Should water and other drinks be included in the guidelines?                                                                                  | ➤ Yes, these should be focused in the national guidelines                                                                                                                                              |
| ➤ Should the guidelines address issues of food and water safety and sanitation, including the safety of complementary foods for young children? | ➤ Yes, it may be given top priority in recommending guidelines for complementary foods for children under five.                                                                                        |
| ➤ Should the guidelines address weight, exercise and other lifestyle factors?                                                                   | ➤ Yes, lifestyle plays a major role in maintaining the body weight, therefore it needs to be included                                                                                                  |

## Steps for Preparing Food Based Dietary Guidelines



## Annex-C

## Pre-testing Questionnaire for National Food Based Dietary Guidelines

|           |                                                                                                                                                                                                                                                            |                                                                     |
|-----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------|
| <b>A</b>  | <b>Demographic Information</b>                                                                                                                                                                                                                             |                                                                     |
| 1.        | Name of Respondent                                                                                                                                                                                                                                         |                                                                     |
| 2.        | Complete Address                                                                                                                                                                                                                                           |                                                                     |
| 3.        | Age (Completed years)                                                                                                                                                                                                                                      |                                                                     |
| 4.        | Sex                                                                                                                                                                                                                                                        | Male Female                                                         |
| 5.        | Education:                                                                                                                                                                                                                                                 | Literate: Yes No Years of Education                                 |
| 6.        | Occupation                                                                                                                                                                                                                                                 | Student / Teacher / House wife / Service / Business / Other Specify |
| 7.        | Family Income (Rupees)                                                                                                                                                                                                                                     | < 10,000 < 20,000 < 30,000 < 40,000 < 50,000 or More                |
| 8.        | Total No of Family Members                                                                                                                                                                                                                                 |                                                                     |
| <b>B.</b> | <b>Knowledge about Food Based Dietary Guidelines</b>                                                                                                                                                                                                       |                                                                     |
| 1.        | Does the given picture of the food guide pyramid provide the information on the messages regarding the use of different food groups in your daily diet?<br><br>If Yes what?                                                                                | Yes<br>No                                                           |
| 2.        | How many food groups are shown in the food guide pyramid?                                                                                                                                                                                                  | Four / Six                                                          |
| 3.        | Do you know that 1 gm protein provides 4 calories, 1 gm carbohydrate provides 4 calories and 1 gm fat provides 9 calories<br><br>(Protein = meat / meat products, egg, nuts and pulses / beans. CHO = sugar, wheat, rice, potatoes etc. Fat= oils / ghee). | Yes / No<br>Don't Know                                              |
| 4.        | Requirement of a normal healthy adult male is 2550 calories / day                                                                                                                                                                                          | Yes / No<br>Don't Know                                              |
| 5.        | Requirement of a normal healthy adult female is 2160 calories / day                                                                                                                                                                                        | Yes / No / Don't Know                                               |
| <b>C.</b> | <b>Awareness About Healthy Diet Pattern / Practices</b>                                                                                                                                                                                                    |                                                                     |
| 6.        | The trend of using large amounts of banaspati or ghee in traditional curry is gradually declining due to awareness.                                                                                                                                        | True / False                                                        |
| 7.        | The trend of eating sugar is also declining due to awareness.                                                                                                                                                                                              | True / False                                                        |

|                                            |                                                                                                                                     |                                                     |
|--------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------|
| 8                                          | Does the seasonal variation affect the meal consumption patterns in your diet?                                                      | Yes / No<br>Some extent                             |
| 9                                          | How many glasses of water are recommended for drinking for a healthy adult per day.                                                 | 2- 4 / 5-7 / 8-10                                   |
| 10                                         | Drink 8 or more glasses of water daily especially in warm climate to avoid dehydration.                                             | True / False                                        |
| 11                                         | Fried foods and sweet meats should be eaten less                                                                                    | True / False                                        |
| 12                                         | Does eating of vegetables (Dark Green Leafy) and legumes base dishes provide fiber and micronutrients (Iron, iodine and vitamin A). | True / False                                        |
| 13                                         | Use vegetable oil for cooking food to reduce the risk of heart diseases                                                             | True / False                                        |
| 14                                         | Vegetable Oil provides same amount of calories as obtained from banaspati ghee                                                      | True / False                                        |
| 15                                         | Include a cup of milk or milk product in your daily diet                                                                            | True / False                                        |
| 16                                         | Use less salt in the daily diet, eat less the foods that are heavily preserved / pickled in salt                                    | True / False                                        |
| 17                                         | One should prefer fresh and homemade foods.                                                                                         | True / False                                        |
| 18                                         | Physical activity on regular basis is necessary for good health                                                                     | True / False                                        |
| <b>D. Awareness About Children's Diet</b>  |                                                                                                                                     |                                                     |
| 19                                         | Breast-feeding should be initiated soon after the childbirth within:                                                                | 30 min - 1 hr.<br>5 -10 hrs<br>2-3 days             |
| 20                                         | Breast milk provides the required calories and protein to the child up to six months of age                                         | True / False                                        |
| 21                                         | Introduce family foods to child at the age of:                                                                                      | 1 yr /2 yrs / 3 yrs                                 |
| 22                                         | Use clean utensils and wash hands before preparing the food and feeding the child to ensure food safety.                            | True / False                                        |
| 23                                         | Approximate cost of menu per person per day                                                                                         | Rs. 10-20 /-<br>Rs. 30-40 /-<br>Rs 50-60 /- or More |
| Name / Signature of Interviewer:           |                                                                                                                                     | Date:                                               |
| Status of Interview: Complete / Incomplete |                                                                                                                                     |                                                     |

DON: 375

## Results of Pre-testing

Table 1: Area / Gender wise Distribution of Subjects

| Location   | No. of subjects | Percent | Total  |
|------------|-----------------|---------|--------|
| Peshawar   | Male 52         | 56.52   | 92     |
|            | Female 40       | 43.48   |        |
| Rawalpindi | Male 64         | 54.24   | 118    |
|            | Female 54       | 45.76   |        |
| Islamabad  | Male 34         | 44.74   | 76     |
|            | Female 42       | 55.26   |        |
| Quetta     | Male 67         | 71.69   | 92     |
|            | Female 25       | 28.41   |        |
| Karachi    | Male 43         | 40.57   | 106    |
|            | Female 63       | 59.43   |        |
| Faisalabad | Male 55         | 56.70   | 97     |
|            | Female 42       | 43.30   |        |
| Lahore     | Male 43         | 42.16   | 102    |
|            | Female 59       | 57.84   |        |
| Total      | Male 358        | 52.42   | N= 683 |
|            | Female 325      | 47.58   |        |

Table 2: Demographic Information

| Subjects                           | Percent |
|------------------------------------|---------|
| <b>Sex</b>                         |         |
| Male                               | 52.42   |
| Female                             | 47.58   |
| <b>Age (years)</b>                 |         |
| < 20                               | 10.69   |
| 21-35                              | 58.27   |
| 35+                                | 31.04   |
| <b>Years of Education</b>          |         |
| Illiterate                         | 2.05    |
| 1-5                                | 1.46    |
| 6-10                               | 16.84   |
| 11+                                | 76.65   |
| <b>Occupation</b>                  |         |
| Students                           | 40.95   |
| Teachers                           | 6.73    |
| Housewife's                        | 24.06   |
| Service                            | 17.42   |
| Business                           | 10.67   |
| <b>Income (Rs)</b>                 |         |
| < 10,000                           | 31.48   |
| < 20,00                            | 31.29   |
| < 30,0000                          | 20.35   |
| < 40,000                           | 6.30    |
| < 50,000                           | 1.76    |
| <b>Family Size</b>                 |         |
| Up To 2                            | 3.07    |
| 3-5                                | 33.24   |
| 6-10+                              | 63.69   |
| <b>Cost of Menu / Day (Rupees)</b> |         |
| 10-20                              | 11.57   |
| 30-40                              | 27.6    |
| 50-60                              | 28.99   |
| 60+                                | 19.03   |

**Table 3      People's Awareness About Food Guide Pyramid**

| Questions                                                                                                                                                           | Issues                                                    | Awareness Levels | Public Messages                                                                                    |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------|------------------|----------------------------------------------------------------------------------------------------|
| Does the given picture of the food guide pyramid provide the required information about the messages regarding the use of different food groups in your daily diet? | Subjects aware of Food Guide Pyramid                      | 42.89 %          | "Awareness about Food Guide Pyramid helps to maintain balanced diet."                              |
| Do you know that 1 gm protein provides 4 calories, 1 gm carbohydrate provides 4 calories and 1 gm fat provides 9 calories                                           | Awareness about calorific values of different food groups | 43.92 %          | "To know about caloric values of different food groups is essential for balanced diet".            |
| Requirement of a Normal Healthy Adult Male is 2550 Calories per day                                                                                                 | Awareness about caloric requirements of male /female      | 41.73 %          | "Awareness about daily caloric requirements will guide dietary intake from different food groups". |

**Table 4      People's Perceptions about Healthy Diet Practices**

| Questions                                                                                                           | Issues                                                                    | Awareness Levels | Public Messages                                                                     |
|---------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------|------------------|-------------------------------------------------------------------------------------|
| The trend of using large amounts of banaspati or ghee in traditional curry is gradually declining due to awareness. | Peoples awareness / trends regarding use of less ghee in cooking foods    | 66.92 %          | "Use of less ghee in cooking will reduce cardiac problems".                         |
| The trend of eating sugar is also declining due to awareness.                                                       | Intake of sugar in daily life.                                            | 61.73 %          | "Use of less sugar will reduce the risk of Diabetes mellitus and Obesity".          |
| Does the seasonal variation affect the meal consumption patterns in your diet?                                      | Awareness about effect of seasonal variations on dietary pattern"         | 39.03 %          | "To consume foods according to seasonal requirements is essential for good health". |
| How many glasses of water are recommended for drinking for a healthy adult                                          | Awareness regarding quantity of water intake per day in routine by adults | 51.94 %          | " For healthy life 8-10 glasses of water should be consumed per day by adults"      |
| Drink 8 or more glasses of water daily especially in warm climate to avoid dehydration                              | More than 8 glasses per day in hot season                                 | 56.93 %          | " Should drink more than 8 glasses of water in hot season to avoid dehydration".    |
| Fried foods and sweet meats should be eaten less.                                                                   | Use of more fried foods and sweet meats is not good for health            | 41.90 %          | "Use of less fried foods / sweet meats lessens cardiac problems and indigestion"    |

**Table 5 Awareness About Children's Diet**

| Questions                                                                                                | Issues                                                        | Awareness Levels | Public Messages                                                                                                 |
|----------------------------------------------------------------------------------------------------------|---------------------------------------------------------------|------------------|-----------------------------------------------------------------------------------------------------------------|
| Breast-feeding should be initiated soon after the childbirth.                                            | Breast milk, as a first food soon after the childbirth.       | 42.12 %          | "Brest feeding is the only choice which ensures child health and increases survival rate".                      |
| Breast milk provides the required calories and protein to the child up to six months of age.             | Period of breast feeding                                      | 36.95 %          | "Breast feeding up to six months provides all the nutrients required for health and optimal growth of a child". |
| Introduction of family foods to child.                                                                   | Time of introductions of family foods to child.               | 45.21 %          | " Introduction of family food at the age of one year is safe along with breast milk".                           |
| Use clean utensils and wash hands before preparing the food and feeding the child to ensure food safety. | Need of cleanliness / hygienic conditions in preparing foods. | 69.71 %          | "Use of clean utensils and washing of hands ensures safety of foods and reduces morbidity".                     |

| Questions                                                                                                                                      | Issues                                                                                                | Awareness Levels | Public Messages                                                                                         |
|------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|------------------|---------------------------------------------------------------------------------------------------------|
| Does eating of vegetables (Dark Green Leafy Vegetables) and legumes base dishes provide fiber and micronutrients (Iron, iodine and vitamin A). | Vegetables are natural source of dietary fiber and micronutrients                                     | 42.36 %          | "Vegetable are source of micronutrients and dietary fiber reduces the risk of constipation and cancers" |
| Use vegetable oil for cooking food to reduce the risk of heart diseases                                                                        | Cooking oil is rich in unsaturated fatty acids instead of ghee containing more saturated fatty acids. | 63.19 %          | " Use of cooking oil instead of ghee reduces risk of heart diseases"                                    |
| Include a cup of milk or milk product in your daily diet                                                                                       | Milk / milk products are (dairy products) good for health and is source of calcium.                   | 43.95 %          | "Milk is good for maintaining health and growth and is a natural source of dietary calcium"             |
| Use less salt in the daily diet, eat less the foods that are heavily preserved / pickled in salt.                                              | Use of salt in daily diet                                                                             | 44.78 %          | "Use less salt in daily diet to reduce risk of hypertension e.g. blood pressure".                       |
| One should prefer fresh and homemade foods.                                                                                                    | Use of fresh / homemade foods                                                                         | 47.07%           | "Fresh and homemade foods are healthy and reduces risk of food contamination and diarrhea".             |
| Physical activity on regular basis is necessary for good health.                                                                               | Doing exercise on regular basis                                                                       | 48.83 %          | Regular exercise reduces risks of heart diseases, diabetes and keep fresh healthy".                     |